Experimental methods in solid-state physics

09 Graphene Band Structure

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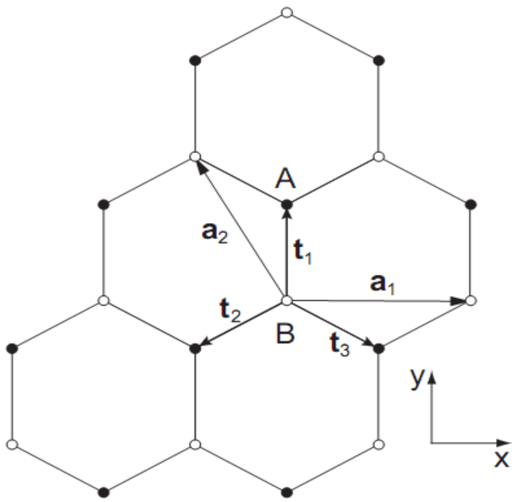
Graphene

Crystal structure:

- Hexagonal lattice
- 2 lattice sites in primitive cell

Real space:

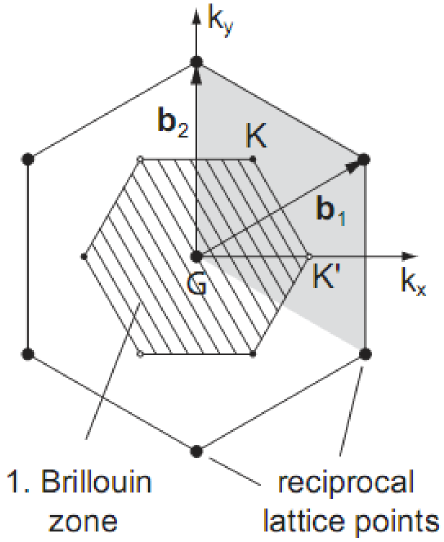


Base vector:

$a_1 = a(1,0)$

$a_2 = a(-1/2, \sqrt{3}\ /2)$

Momentum space:



1^{st} Brillouin zone: two lattice sites in primitive unit cell (real space)

→ K and K' different lattice sites in BZ

→ Valley degree of freedom

Tight binding approach

<u>Assumption for tight binding approximation (TBA):</u>

1. Energy eigenvalues and eigenfunctions are known for an electron in an isolated atom.
2. When atoms are brought together to form a solid, they remain sufficiently far apart so that the electron can be assigned to a particular electronic state.
3. Periodic potential is approximated by a superposition of atomic potentials.
4. Perturbation theory can be used to treat the difference between actual potential and the atomic potential.

Perturbation is the difference between the periodic potential and the atomic potential, around which the electron is localized.

General idea of TBA is to construct a trial wave function ψ from the (known) orbital wave function ϕ_i.

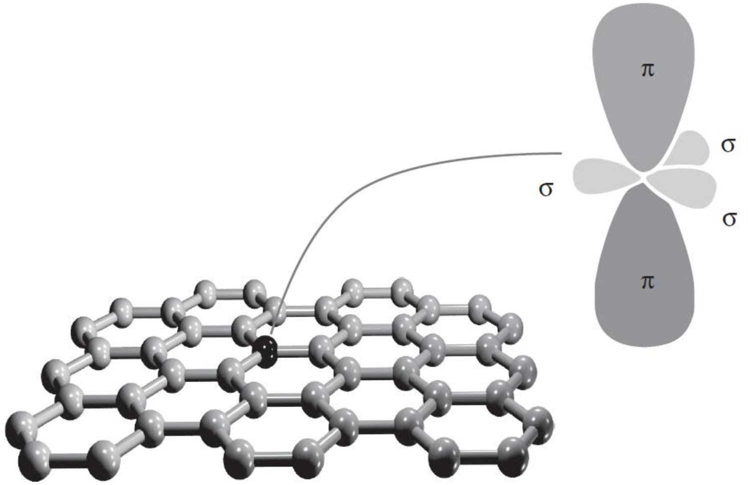
Bonding in graphene

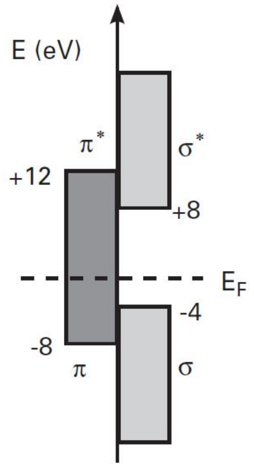
Electron configuration of C-atom:

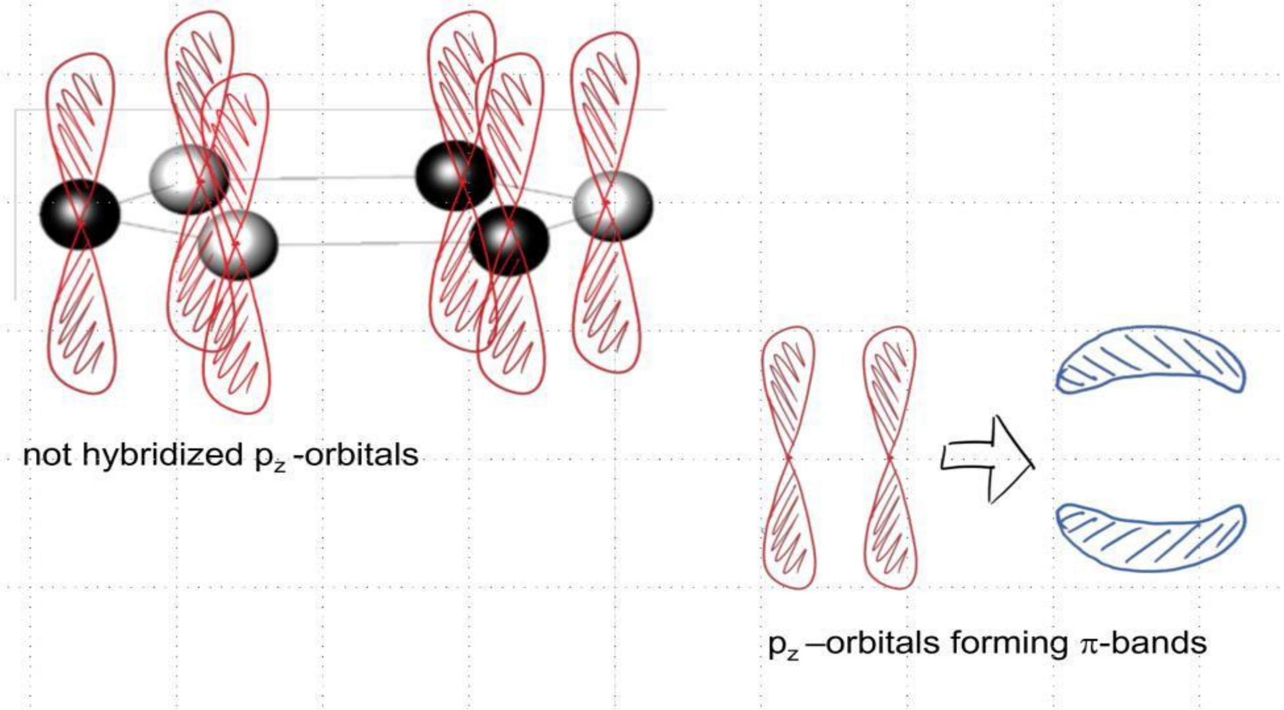
$1s^2 2s^2 2p^2 \equiv [He]\ 2s^2 2p^2$ → **sp^2-hybridisation:**

- 4 valence electrons per C-atom
- 8 valence electrons per unit cell in graphene
- *sp^2*-orbitals form strong σ-bonds (6 valence electrons)
- p_z – orbital perpendicular to plane of hybridization (2 valence electrons)
- p_z – orbitals overlap and form weak π-bonds

p_z -orbitals determine electronic band structure in graphene







L.E.F. Foa Torres, Introduction to Graphene-Based Nanomaterials, Cambridge (2014).
available as ebook at https://opac.ub.tum.de

Electronic band structure of graphene

- p_z– orbitals centered at lattice site A and B
- p_z– orbitals only weakly coupled in x-y direction

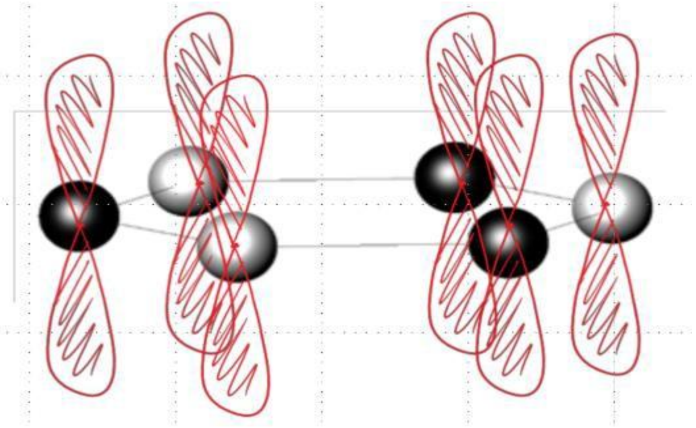
→ binding and anti-binding state

- sp^2-hybridized orbitals strongly bound

→ 6 valence electrons strongly localized

→ do not contribute to conductivity of graphene

→ can be neglected in the electronic band structure calculation



Tight Binding ansatz:

ϕ_A, ϕ_B: p_z– orbitals on lattice site 1 and 2

Summation:

(1) $$\phi(\vec{x}) = c_A\phi_A(\vec{x}) + c_B\phi_B(\vec{x})\Sigma_M = \sum_{\mathrm{n=A,B}} C_{\mathrm{n}}\phi_n(\vec{x})$$

Hamiltonian of electron in periodic potential (in agreement with Bloch Ansatz):

(2) $$H = \frac{\vec{p}^2}{2m} + \sum_{\mathrm{i}}(V\left(\vec{x} - \vec{x}_A - \vec{R}_i\right) + V\left(\vec{x} - \vec{x}_B - \vec{R}_i\right))$$

x_j:: vector joining a lattice point and the second atom in the basis.

Application onto wave function ϕ_i:

(3) $$H\phi_A = \varepsilon_A \phi_A + \underbrace{\left(\frac{\vec{p}^2}{2m} + \sum_i (V(\vec{x} - \vec{x}_A - \vec{R}_i) + V(\vec{x} - \vec{x}_B - \vec{R}_i))\right)}_{\Delta U_1} \phi_A$$

- analogue for lattice site B
- ε_i: eigenvalues of p_z-orbital at lattice site *i*
- ΔU_A small around lattice site *B*
- ϕ_i small away from lattice site *i*

for $\varepsilon_A = \varepsilon_B = 0$:

(4) $$H\phi_i = \Delta U_i \phi_i$$

(5) **Schrödinger equation:** $$H\psi_{\vec{k}} = E(\vec{k})\,\psi_{\vec{k}}$$

Projection of ϕ_1 and ϕ_2 on wave function ψ (to get c_1 and c_2 from equation (1))

(6) $$E(\vec{k})\langle \phi_i | \psi \rangle = \langle \phi_i | \Delta U_i | \psi \rangle$$

(6) $E\left(\vec{k}\right)\langle\phi_i|\psi\rangle = \langle\phi_i|\Delta U_{\mathrm{i}}|\psi\rangle$

with overlap integral of nearest neighbors:

(7) $\langle\phi_{\mathrm{A}}|\psi\rangle = c_{\mathrm{A}} + c_{\mathrm{B}}\left(\int\phi_A * \phi_B\right)\left(1 + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{A}}}} + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{B}}}}\right)$

$\langle\phi_{\mathrm{B}}|\psi\rangle = c_{\mathrm{B}} + c_{\mathrm{A}}\left(\int\phi_A * \phi_B\right)\left(1 + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{A}}}} + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{B}}}}\right)$

Next, with following definitions:

(8) $\gamma_0 = \int\phi_A * \phi_B \quad \epsilon\mathbb{R}$ — overlap integral (nearest neighbor)

(9) $\alpha\left(\vec{k}\right) = 1 + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{A}}}} + e^{-\vec{k}\,\overrightarrow{a_{\mathrm{B}}}}$ — phase factor

(10) $\gamma_A = \int\phi_A * \Delta U_A\phi_B = \int\phi_B * \Delta U_B\phi_A \quad \epsilon\mathbb{R}$ — transfer integral (nearest neighbor hopping amplitude)

(6) ➡

$$\begin{pmatrix} E\left(\vec{k}\right) & \alpha\left(\vec{k}\right) * \left(\gamma_0 E\left(\vec{k}\right) - \gamma_1\right) \\ \alpha\left(\vec{k}\right) * \left(\gamma_0 E\left(\vec{k}\right) - \gamma_1\right) & E\left(\vec{k}\right) \end{pmatrix}\begin{pmatrix} c_{\mathrm{A}} \\ c_{\mathrm{B}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} E(\vec{k}) & \alpha(\vec{k}) * (\gamma_0 E(\vec{k}) - \gamma_1) \\ \alpha(\vec{k}) * (\gamma_0 E(\vec{k}) - \gamma_1) & E(\vec{k}) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

➡ **from Determinant:** $\boxed{E(\vec{k}) = \pm\gamma_0 \left|\alpha(\vec{k})\right| + \gamma_1 \left|\alpha(\vec{k})\right|^2}$

(for graphene: $\gamma_1 \to 0$ because of vanishing overlap in x-y direction)

Relation is symmetric around zero (for $\gamma_1 \to 0$)

➡ **Dispersion relation of graphene:** $\boxed{E(\vec{k}) = \pm\gamma_0 \left|\alpha(\vec{k})\right|}$

"+": conduction band
"-": valence band

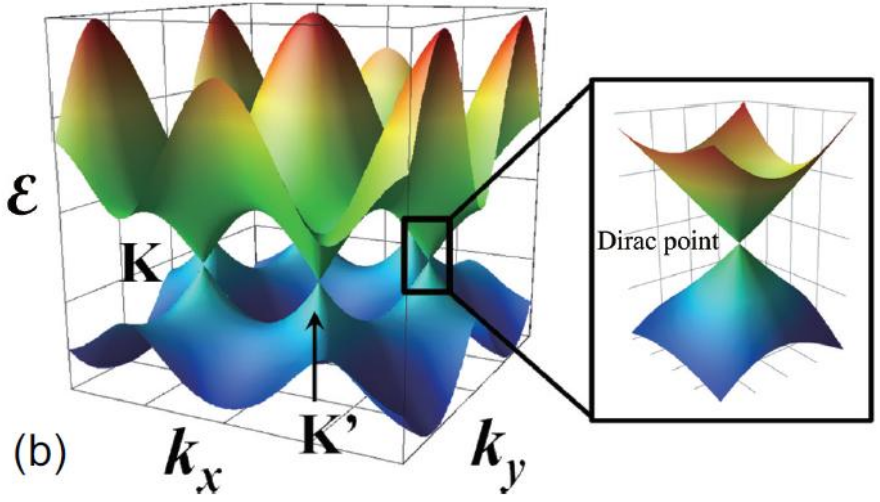
for x- and y- components of vector *k*,

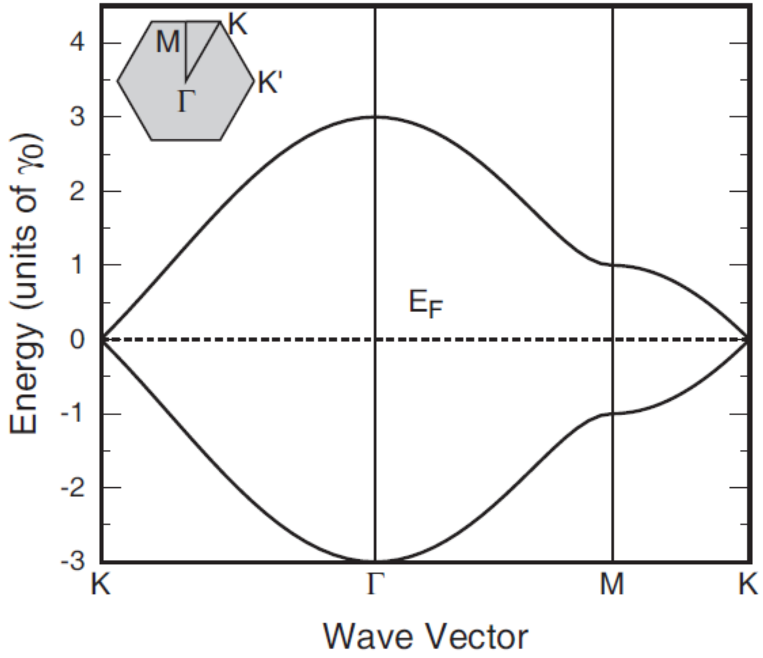
- *together with $e^{ix}+e^{-ix} = 2\cos(x)$*
- *trigonometric identities*
- *primitive lattice vector*

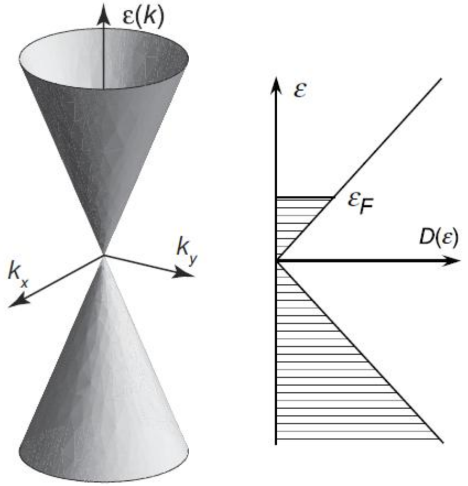
$$\alpha(\vec{k}) = \left[\left(1 + e^{\vec{k}\vec{a}_A} + e^{\vec{k}\vec{a}_B}\right)\left(1 + e^{-\vec{k}\vec{a}_A} + e^{-\vec{k}\vec{a}_B}\right)\right]^{1/2}$$
$$= \cdots$$
$$= \left[3 + 2\cos\left(\sqrt{3}k_y a\right) + 4\cos\left(\frac{\sqrt{3}}{2}k_y a\right)\cos\left(\frac{3}{2}k_x a\right)\right]$$

Dispersion relation of graphene

$$E(k_x,k_y)=\pm\gamma_0\sqrt{1+4\cos\left(\frac{\sqrt{3}ak_y}{2}\right)\cos\left(\frac{ak_x}{2}\right)+4\cos^2\left(\frac{ak_x}{2}\right)}$$







Hexagonal structure in 1st BZ with linear dispersion

Figs. 2.2, 2.3, 3.22 in Toshiaki Enoki, Tsuneya Ando „Physics and Chemistry of Graphene“
Jenny Stanford Publishing 2019, available as ebook at https://opac.ub.tum.de

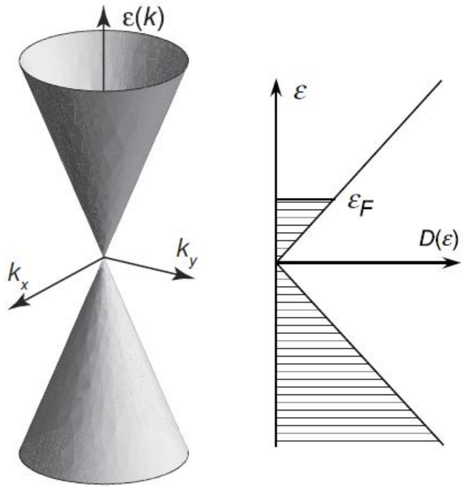
Dispersion relation of graphene

Peculiarities:

• Half-metallic behavior
→ *with Fermi-energy E_F continuously tunable from valence to conduction band*

• Linear Dispersion at charge neutrality points *K*, *K*‘ (Dirac Points)
→ *charge carriers behave like massless fermions (Dirac Fermions)*
with Fermi velocity $v_F \sim 10^6$ m/s

• Relevant quantum numbers for charge carriers:
→ Spin (up and down)
→ Pseudospin (K or K’ valley)



Quantum mechanical (low-energy / toy model) description of valley degree of freedom:

$$\boxed{H = \begin{pmatrix} 0 & k_x - i k_y \\ k_x + i k_y & 0 \end{pmatrix} = \hbar v_F \vec{\sigma}\vec{k}}$$

Pseudospin described by 2-dimensional Pauli-Spin-Matrices σ

Photoelectric effect

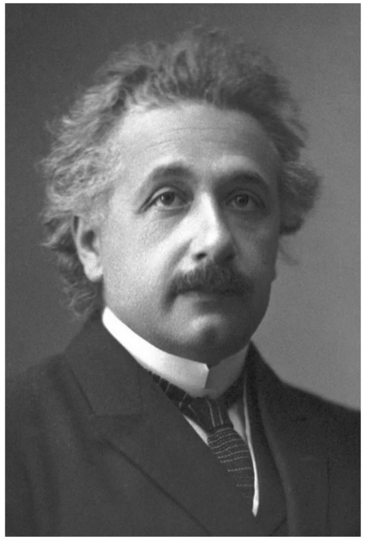
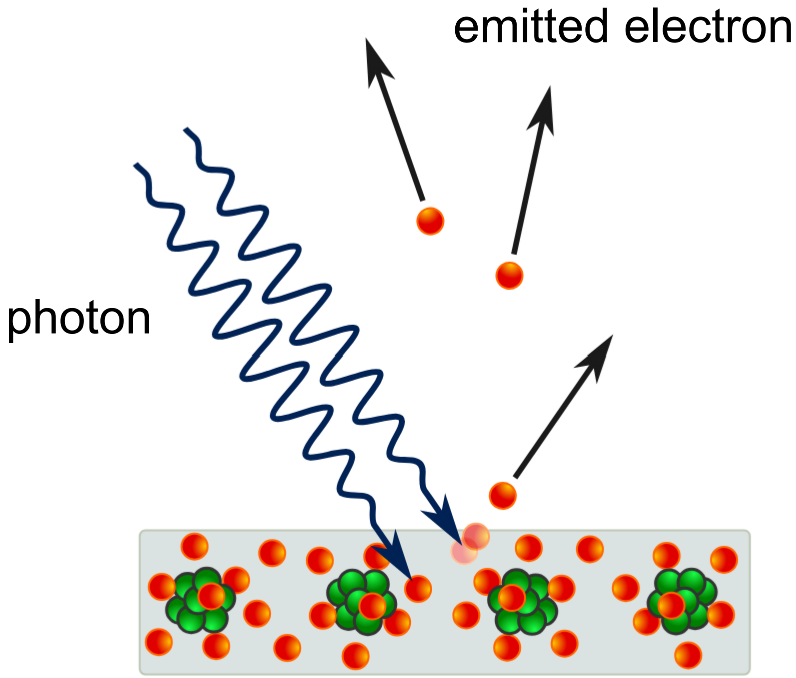
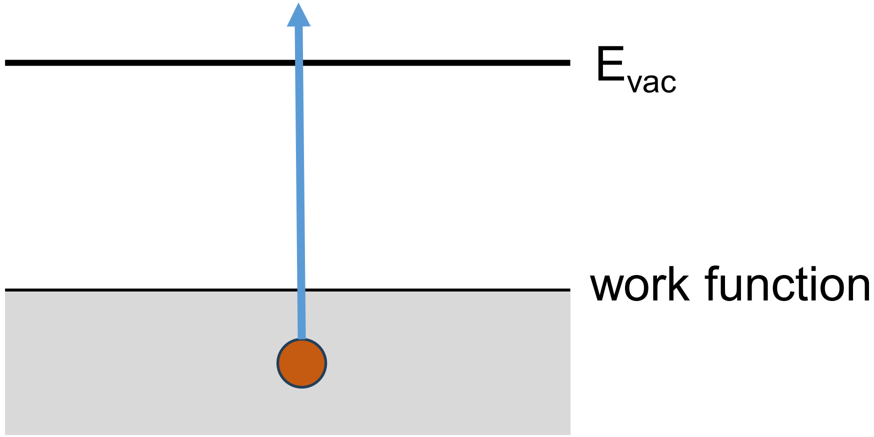


Photo from the Nobel Foundation archive.

Albert Einstein

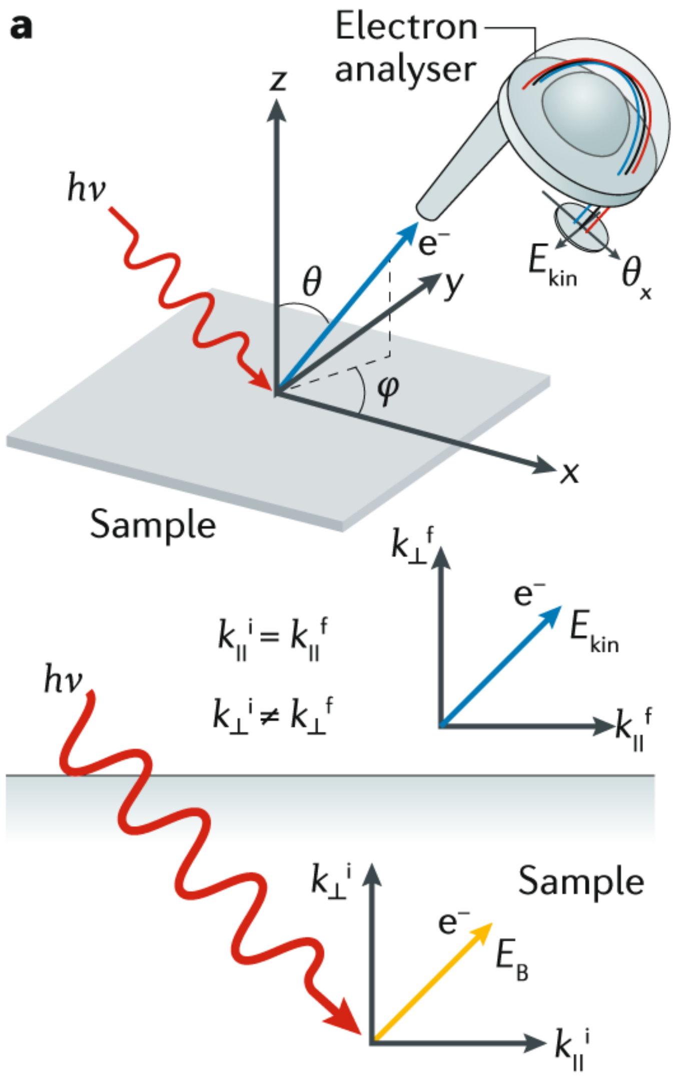
Prize share: 1/1

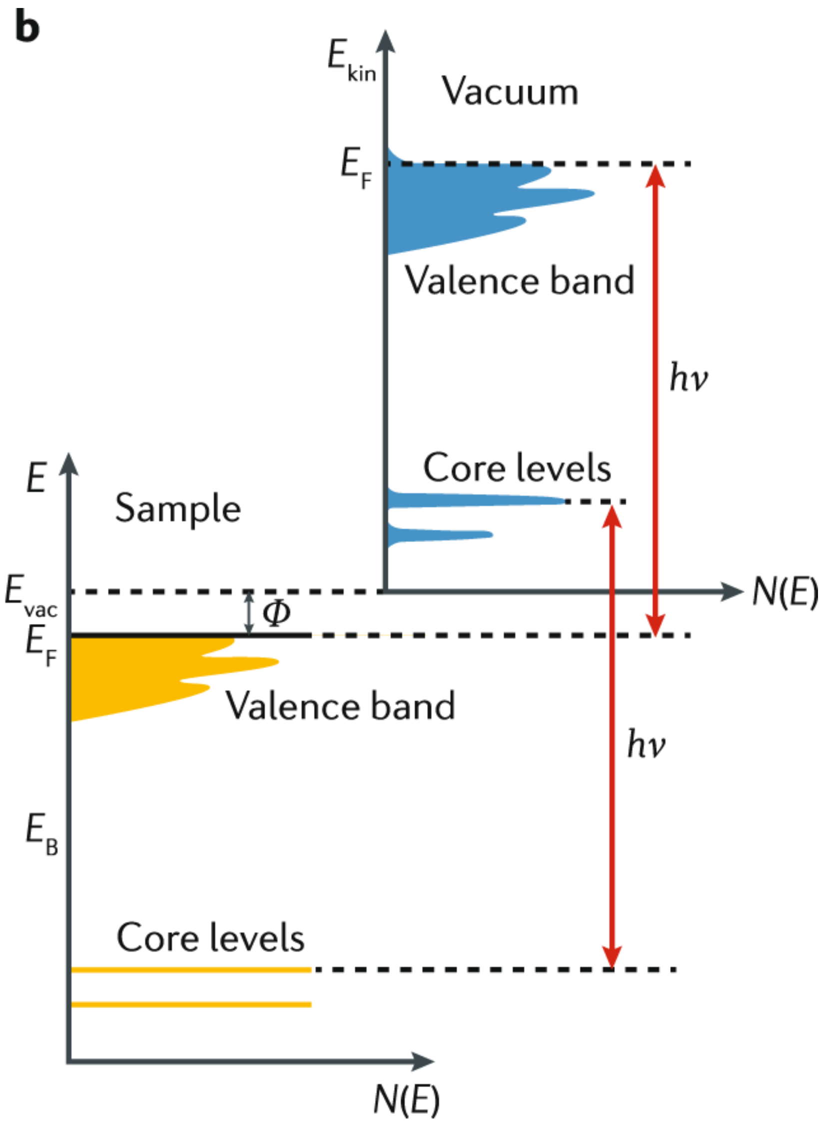




The Nobel Prize in Physics 1921 was awarded to Albert Einstein "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

ARPES – Measuring band structure directly





https://www.nature.com/articles/s42254-019-0088-5

Energy conservation

$$E_{kin} = h\nu - \phi - E_B$$

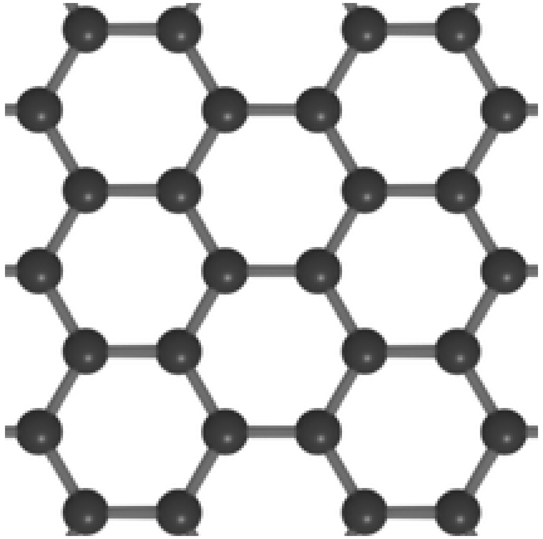
In-plane momentum conservation

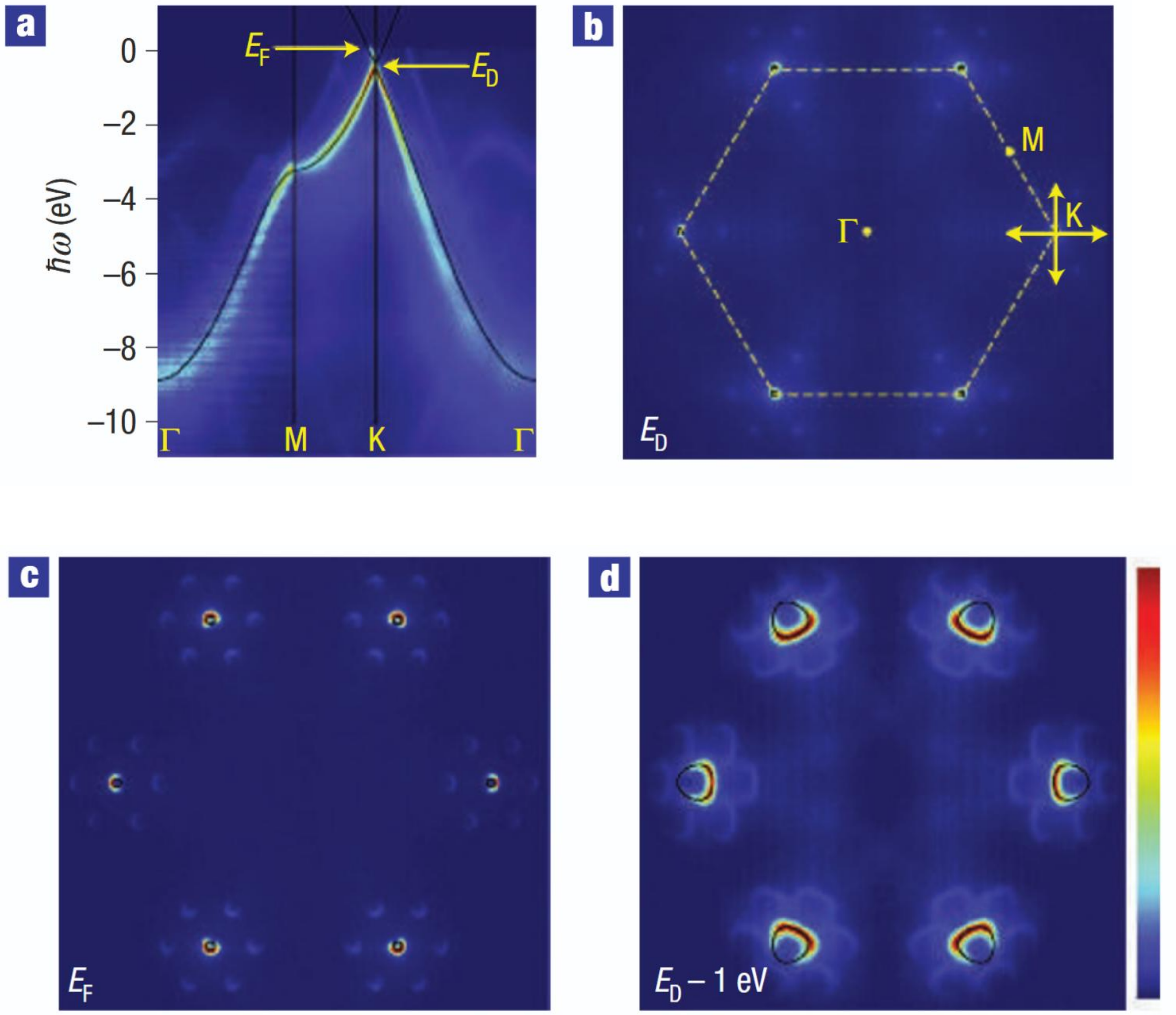
$$\hbar k_{i,\parallel} = \hbar k_{f,\parallel} = \sqrt{2mE_{kin}}(\sin(\theta)\cos(\phi)k_x + \sin(\theta)\sin(\phi)k_x)$$

Out-of-plane momentum (potential step at surface)

$$\hbar k_\perp = \sqrt{2m(E_{kin}\cos^2(\theta) + V_0}$$

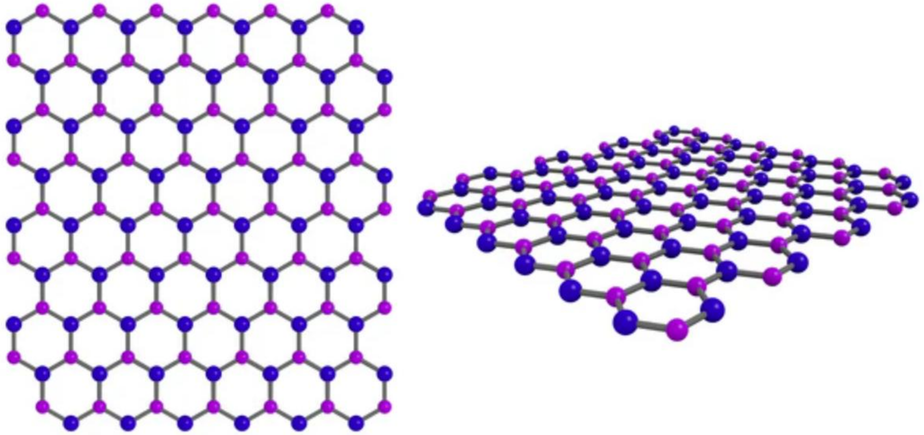
ARPES – graphene

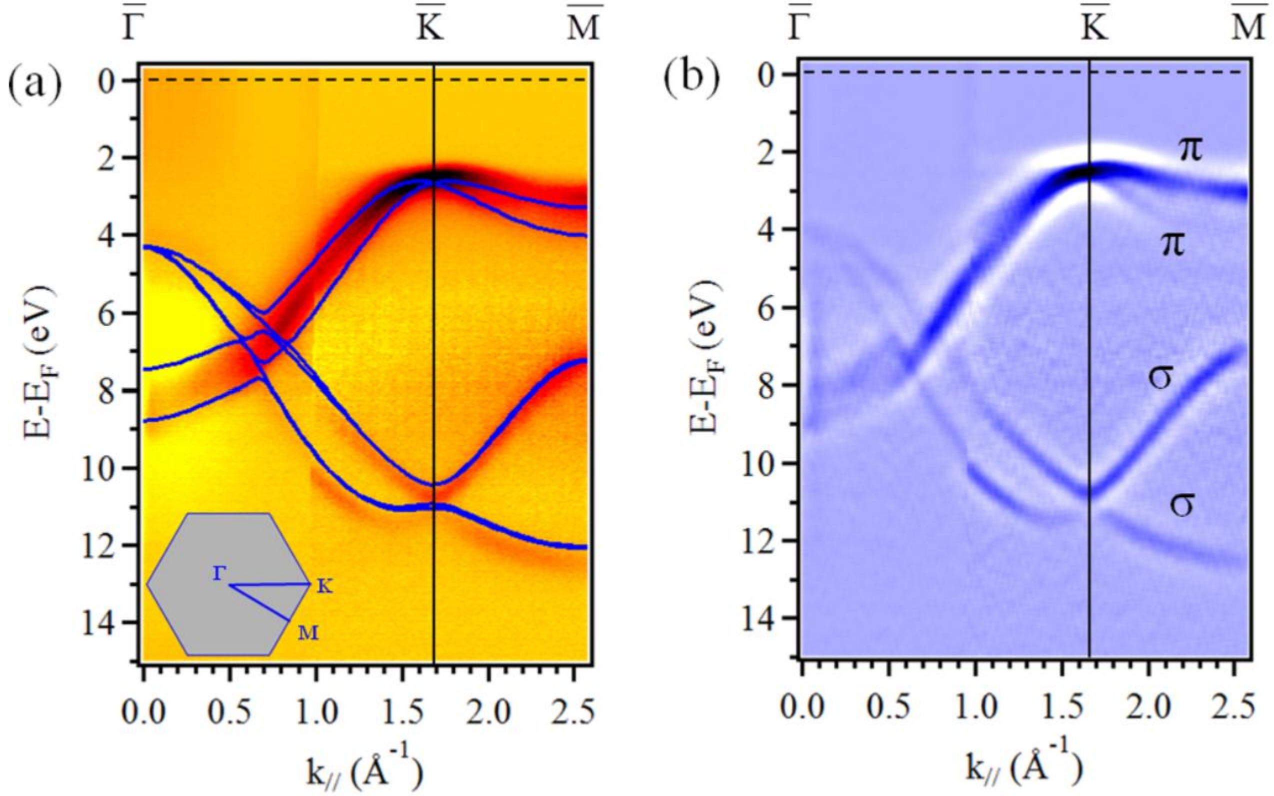




https://www.nature.com/articles/nphys477

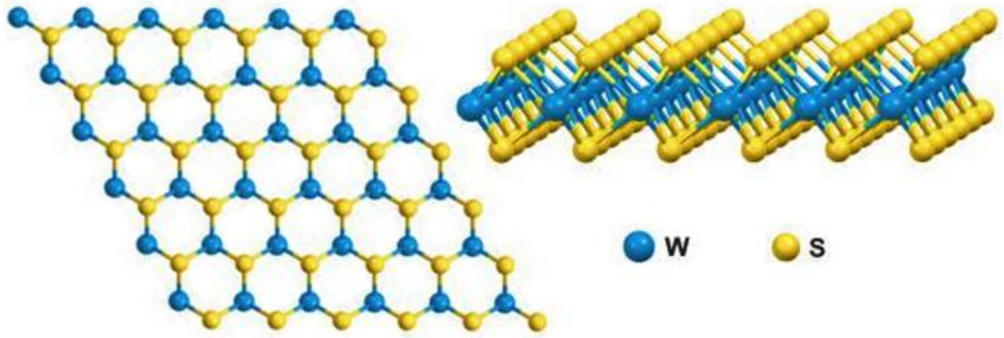
ARPES – hexagonal boron nitride (hBN)

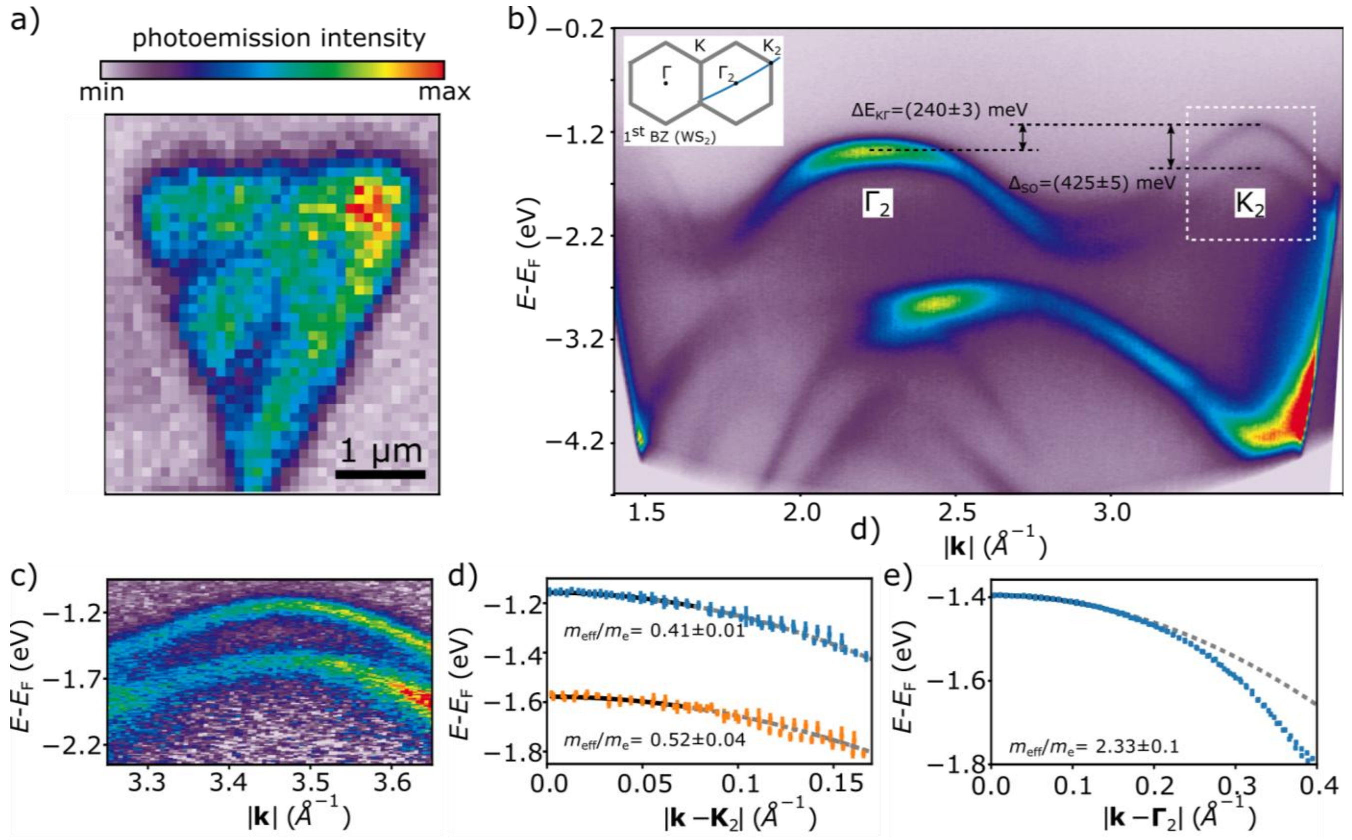




https://journals.aps.org/prb/pdf/10.1103/PhysRevB.95.085410

ARPES – tungsten disulfide (WS_2)





10.1088/2053-1583/aad21c